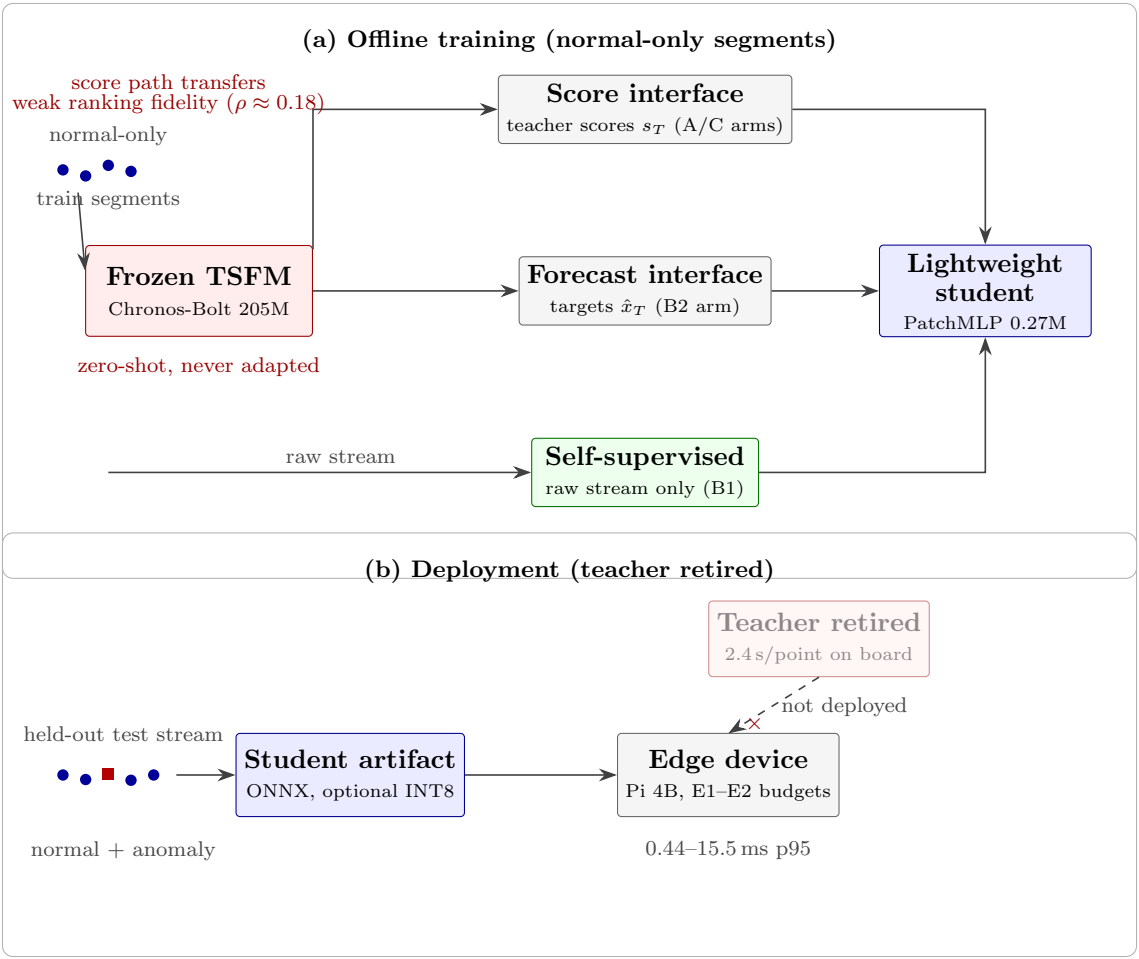

(a) Offline training (normal-only segments)
score path transfers
weak ranking fidelity ($\rho \approx 0.18$)
normal-only
train segments
Score interface
teacher scores s_T (A/C arms)
Frozen TSFM
Chronos-Bolt 205M
zero-shot, never adapted
Forecast interface
targets $\hat{x}_T$ (B2 arm)
Lightweight student
PatchMLP 0.27M
raw stream
Self-supervised
raw stream only (B1)
(b) Deployment (teacher retired)
Teacher retired
2.4 s/point on board
not deployed
held-out test stream
Student artifact
ONNX, optional INT8
Edge device
Pi 4B, E1–E2 budgets
normal + anomaly
0.44–15.5 ms p95